virtual machines known as nodes. A pod is itself a collection of Linux containers, each of which typically has a single process running within it, such as an API server.

A Linux container is the name given to a collection of technologies within the Linux operating system that allow a process (or collection of processes) to be isolated from other processes so that it sees its own view of the file system, network, users, and other shared resources. This simplifies packaging and deployment, because different processes can use different versions of the same components, which might otherwise cause conflicts. You can even run entirely different distributions of Linux within containers simultaneously on the same operating system kernel. Containers also provide security benefits, because processes can be locked down within a container such that it is much harder for an attacker that compromises one process to break out of the container and affect other processes running in different containers or the host operating system. In this way, containers provide some of the benefits of VMs, but with a lower overhead. Several tools for packaging Linux containers have been developed, the most famous of which is Docker (https://www.docker.com), which many Kubernetes deployments build on top of.

> LEARN MORE Securing Linux containers is a complex topic, which we'll cover only the basics of in this book. The NCC Group have published a freely available 123-page guide to hardening containers (https://www.nccgroup.trust/us/our-